Worksheet 1-1 A Change at an Instant #1
AP Calculus Name_____

1.) A candle is burning in a quiet room. The height of the candle, $H(t)$, in centimeters, is measured at various times t, where t is the number of minutes since it was lit. The table below shows the recorded height at selected times.

t (minutes)	0	3	7	11
$H(t)$ (cm)	12.0	11.6	10.9	10.2

a.) Find the average rate of change in the height of the candle between $t = 0$ and $t = 11$. Include units in your answer.

b.) Estimate the instantaneous rate of change in the candle's height at $t = 5$ minutes. Show your reasoning and include units in your answer.

2.) The number of customers at a coffee shop t hours after it opens can be modeled by $C(t) = -2(t-5)^2 + 50$ where $0 \leq t \leq 10$.

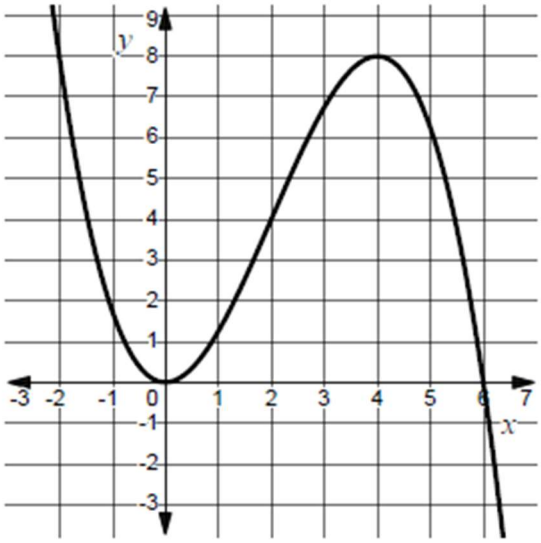
a.) Find the average rate of change in the number of customers between $t = 1$ and $t = 8$.

b.) Use the function to calculate the average rate of change on the intervals [3, 3.5], [3, 3.1], [3, 3.01] and [3, 3.001].

c.) We can calculate the instantaneous rate of change at $t = 3$ by calculating the average rate of change over a very small interval around 3. What do you notice about the values of the average rate of change as the interval around 3 gets smaller? What do you think is the instantaneous rate of change at $t = 3$?

3.) The graph of $y = f(x)$ is shown.

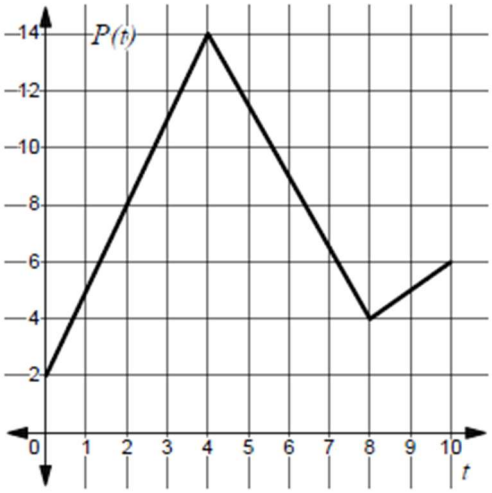
a.) Is the instantaneous rate of change of f greater at $x = 2$ or $x = 3$? How do you know?



b.) Identify one x-value where the instantaneous rate of change is 0.

c.) Identify one interval where the average rate of change of f is 0.

4.) A robot vacuum moves back and forth along a long straight hallway. The robot's position, measured in meters from the charging station, is given by $P(t)$ where t is the time in minutes. The graph of P is shown for $0 \leq t \leq 10$.



a.) Determine the instantaneous rate of change in the robot's position at $t = 5$.

b.) What does your answer in part (a) indicate about the robot's movement?

5.) Consider the graph of $f(x) = (x - 2)^2 - 5$. Order the following from least to greatest.

I. The average rate of change of f on the interval [–1, 5].
II. The instantaneous rate of change of f at $x = 10$.
III. The instantaneous rate of change of f at $x = 8$.
IV. The average rate of change of f on the interval [2, 8].
V. The instantaneous rate of change of f at $x = 0$.

6.) The number of cans of soup produced at a factory can be modeled by a function C, where $C(t)$ gives the number of cans produced t hours after the factory begins their production lines at 5 AM. Which of the following is the best estimate of the instantaneous rate of change in the factory's soup can production at 7 AM?

A.) $\frac{C(2)-C(2)}{2-2}$

B.) $\frac{C(7)}{7}$

C.) $\frac{C(2.1)-C(1.9)}{0.2}$

D.) $\frac{C(3)-C(1)}{2}$

7.) Let g be the function given by $g(x) = \sqrt{x} - 3x$. Find the average rate of change of g on the closed interval [4, 9].

A.) $-\frac{34}{5}$

B.) $-\frac{14}{5}$

C.) $-\frac{8}{3}$

D.) 2